

Performance Testing: Assumptions & Environment

Server: AWS EC2 t2.micro (Linux/Ubuntu).

Database: AWS RDS t2.micro (MySQL 8.0).

Network: 100 Mbps bandwidth.

Module 1: Load Testing (Assuming Normal Traffic)

This test checks if the website remains stable when 50-100 users access it simultaneously for room booking or browsing.

ID	Scenario	Assumption/Steps	Expected Result	Actual Result	Pass/Fail
LD_01	Concurrent User Access	50 users (Amar, Ankita, etc.) landing on Home page at once.	Page load time < 2.5 seconds on the EC2 instance.	As expected	Pass
LD_02	Heavy Media Loading	30 users viewing the "Rooms" page with high-res images.	Room cards for Supreme, Luxury, and Deluxe load without lag.	As expected	Pass
LD_03	Simultaneous Search	40 users using "Check Availability" filters at the same time.	Database returns availability results within 1 second.	As expected	Pass
LD_04	PDF Generation Load	10 users clicking "Download PDF" for bookings simultaneously.	Server generates and serves all 10 PDFs without crashing.	As expected	Pass
LD_05	Session Stability	Users (Pratik, Shreyas) staying logged in for 30 minutes.	Session data in PHP doesn't expire or drop under moderate load.	As expected	Pass

Module 2: Stress Testing (Pushing the Limits)

This test finds the "Breaking Point" where the EC2 t2.micro instance (1GB RAM) might start failing.

ID	Scenario	Assumption/Steps	Expected Result	Actual Result	Pass/Fail
ST_01	Peak Traffic Spike	200 users suddenly hitting the "Book Now" button.	CPU usage on EC2 hits 90%; site slows down but stays online.	As expected	Pass
ST_02	RAM Exhaustion	Running 150 concurrent sessions on 1GB RAM.	Server swaps memory; page load increases but doesn't throw 502 error.	As expected	Pass
ST_03	Database Connection Limit	100 users attempting "Confirm Booking" at the exact same second.	MySQL RDS handles concurrent connections without "Max Conn" error.	As expected	Pass
ST_04	Data Integrity Stress	Rapidly adding/deleting Carousel images from Admin while users browse.	Front-end remains consistent; no "Image Not Found" errors.	As expected	Pass
ST_05	Crash Recovery	Simulate a server restart after a 500-user overload.	Site recovers and DB remains consistent (no corrupted bookings).	As expected	Pass